



Tribhuvan University
Faculty of Humanities and Social Sciences
Ratna Rajyalaxmi Campus
Mid-Term Examination 2024

Class R.N.

Bachelor in Computer Application (BCA)
Course Title: Operating System
Code No: CACS 251
Semester: IV

Full Marks: 40
Pass Marks: 16
Time: 2 hours

Candidates are required to answer the questions in their own words as far as possible

Group B

Attempt any FOUR questions.

[4*5=20]

1. Explain Operating system as a Extended machine and resource manager.
2. What is multi-threading OS? How multithreading improves performance over an single-threaded solution?
3. Differentiate between process and thread.
4. How does semaphore provide the solution to the producer-consumer problem?
5. How multithreading improves performance over an single-threaded solution?
6. Differentiate between preemptive and non-preemptive scheduling.
7. Explain Peterson's Algorithms and TSL Hardware solutions in detail.
8. Write short note on PCB.

Group C

Attempt any ONE question.

[1*10=10]

9. What is Kernel? What role does it plays in an Operating System? Differentiate between monolithic and micro kernel.
10. Explain the idea of Reader writer problem and sleeping barber problem and solve it using Semaphore.
11. Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	5
P4	1	4
P5	5	2

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

- a. Draw four Gantt charts illustrating the execution of these processes using FCFS, SJF, a non-preemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1) scheduling.
- b. Find average turn around time and wating time for each scheduling algorithm.
- b. What is the turnaround time of each process for each of the scheduling algorithms in part a?
- c. What is the waiting time of each process for each of the scheduling algorithms in part a?

Best of Luck !

non-preemptive is not Flexible



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Mid-Term Examination 2025

Bachelor in Computer Application (BCA)
Course Title: Scripting Language
Code No: CACS 254
Semester: 4th

Full Marks: 40
Pass Marks: 16
Time: 2 hours
Symbol No:

Candidates are required to answer the questions in their own words as far as possible

Group B

Attempt any FOUR questions.

[4×5= 20]

1. How client side scripting and server side scripting are different? Explain. [5]
2. Explain about event handling in JavaScript. Discuss any two ways to register event handler with appropriate examples. [2.5 + 2.5]
3. What is DOM manipulation? Explain any three DOM manipulation tasks using jQuery. [2 + 4]
4. What is an associative array in PHP? Write a program to demonstrate the concept of 2 D associative array in PHP with programming example. [2+ 3]
5. Create a html form having two text field and a submit button. Now write PHP code to take two numbers from the text fields and display "Happy" if the sum of two numbers is divisible by 5, display "Fit and Fine" if the sum is divisible by 3, display "Programming is Fun" if the sum is greater than thrice the second number and display "Yes!Yes!!" otherwise. [5]

Group C

Attempt any ONE question

[1×10 = 10]

1. Explain about the importance of regular expressions. Create a web application having an image, when mouse cursor is placed over the image, description of the image should be displayed on a paragraph below that image and when muse cursor is removed from the image, the description should be vanished. Now place a form below that with validation using JS. The validation errors should be displayed after each field if invalid considering following validation rules. [2+ 3 + 5]
 - All fields are required. (name, username, age, password and confirm password)
 - Name cannot have numbers and special characters
 - Username should be of minimum 8 character and must begin with alphabet
 - Age cannot be less than 18 years
 - Password and confirm password must match.
2. Create a simple CRUD application using PHP and MySQL. Make your own assumption about database and table. The input field must be validated in server side. [10]

Group B

[5 × 2] = 10

Attempt any two questions

11. Applying Gauss-Seidal iteration method to solve the system of equations $10x + y - z = 11$, $x + 10y + z = 28$, $-x + y + 10z = 35$ correct to two decimal places

12. Applying Lagrange's polynomial formula, y for $x = 2.5$ from the following data

$x:$	1	3	4
$y:$	5	9	11

13. Applying Cubic spline interpolation method to find $y(7.5)$

$x:$	2	4	6	8
$y:$	10	14	15	18

14. Fitting in exponential function $y = a b^x$ from the following data

x	2	4	6	8	10	12
y	16	11.1	8.7	6.4	4.7	2.6

15. Factorize the matrix $A = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 5 & 2 \\ 2 & 1 & -3 \end{bmatrix}$ applying LU decomposition method

16. Solve the non-linear equation $x e^x - \cos x = 0$ using Newton-Raphson method

Group C

[10 × 1 = 10]

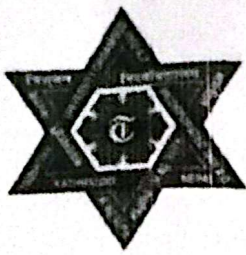
Attempt any one question

17. Find an approximation root of non-linear equation $x^3 - 4x - 9 = 0$ correct to three decimal places with initial guess $a = 2$ and $b = 3$

18. Using cubic spline technique, estimate $y(9)$ from the following data

x	4	6	8	10
y	2	5	8	6

19. Applying Gauss-Jordan method to solve the system of equations $3x - 2y + z = 0$, $x - 3y - 5z = -19$, $2x - y + 3z = 9$



Tribhuvan University
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Pradarshani Marga, Kathmandu
Mid Term Examination 2082

Bachelor in Computer Application (BCA)
Course Title: Software Engineering
Code No: CACS253
Semester: IV

Full Marks: 40
Pass Marks: 16
Time: 2 hours

Candidates are required to answer the questions in their own words as far as possible.

Group – 'B'

Attempt any FOUR questions.
5 = 20]

[4 x

2. What are the differences between software engineering and system engineering? Explain.
3. What is a prototype model? Under what circumstances it is beneficial to construct a prototype model.
4. Why functional and non-functional requirements must be considered together while preparing software specification? List some of the functional requirements and non-functional requirements for "Online Library Management System" of a college.
5. Explain RUP framework in detail.
6. What do you understand by CASE? Explain the types of CASE.

Group – 'C'

Attempt any ONE Question.

[1 x 10 = 10]

7. What is requirements elicitation? What are the different types of requirements elicitation techniques? Explain.
8. Explain Requirements Engineering Process (REP) and its phases along with a proper figure.

Best of Luck!!!



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Mid-Term Examination 2025

Bachelor in Computer Application (BCA)
Course Title: Database Management System
Code No: CACS 255
Semester: V

Full Marks: 40
Pass Marks: 16
Time: 2 hours

Group B

Attempt any six. [5 X 6 = 30]

2. What is data model? What are the types of data model?
3. What is database? What are the merits and demerits of database management system?
4. Describe data independence with its types.
5. Write a procedure to convert ER diagram to relation.
6. What is stored procedure? How it differs from trigger?
7. What is DDL and DML queries? Consider the following relation,

customer name	loan number
X	L01
Y	L02

Relation borrower

loan number	branch name	amount
L01	B1	5000
L02	B2	6000

Relation loan

Write a relational algebra expression to find all customer who taken loan from branch "B1".

8. Describe the pitfalls of relational database design.

Group C

Attempt any two [2 X 10 = 20]

9. What is three schema architecture of database. Describe database system architecture.
10. Describe the importance of normalization in DBMS. Describe 1NF, 2NF, 3NF, BCNF and 4NF with examples.
11. Consider following tables.
Shoppe(id, sname, area)
Accessories(no, name, price, id)
Write a sql queries for following
 - i. To display name and price of all accessories in descending order of their price.
 - ii. To display name, price of all accessories and their respective sname where price is 5000 or more.
 - iii. To display name, minimum and maximum price of each name from accessories table.
 - iv. To display all details of accessories where name contains word "Board".